

PYTHON BASIC

BY AMAR PANCHAL

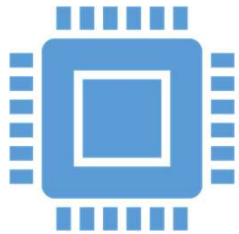
PYTHON IS

- Python is a high-level, interpreted, interactive and object-oriented scripting language. Python was designed to be highly readable which uses English keywords frequently whereas other languages use punctuation and it has fewer syntactical constructions than other languages. Created by Guido van Rossum and first released in 1991

History of Python

- December 1989 | Implementation by Guido van Rossum as successor of ABC to provide exception handling
- February 1991 | Python 0.9.0 had classes with inheritance, exception handling, functions, and the core datatypes
- January 1994 | Version 1.0 has functional programming features
- October 2000 | Python 2.0 brings garbage collections
- Python 2.2 improves Python's types to be purely object oriented
- December 2008 | Python 3 | Reduce feature duplication by removing old ways of doing things.
- **Not backwards compatible with Python 2.x**

History of Python



Modules

Reusable pieces of software

Can be written by any Python developer

Extend Python's capabilities



Python Web site at www.python.org

Primary distribution center for Python source code, modules and documentation

VERSION PROBLEM

PYTHON 2.X	PYTHON 3.X
started in 2000 and slated to lose all support in 2020	Started in 2008 and seen as future of Python
Absence of new modules	Newer modules like <code>__future__</code> and <code>asyncio</code>
<code>print</code> keyword	<code>print()</code> is a function
ASCII and UNICODE support Use <code>u</code> for Unicode <code>u 'Hello'</code> else byte sequence	UNICODE default All strings are now Unicode <code>b</code> used for byte sequence
Division With Integers $5 / 2 = 2$	Division With Integers $5 / 2 = 2.5$
<code>raw_input()</code> function and an <code>input()</code>	Only <code>input()</code>

PYTHON IS GREAT BECAUSE...



Portable

Your program will run if the interpreter works and the modules are installed



Efficient

Rich language features built in
Simple syntax for ease of reading
Focus shifts to “algorithm”



Flexible

Imperative
Object-oriented
Functional

PYTHON IS GREAT BECAUSE...



Extendible

Plenty of easy-to-use modules

If you can imagine it, then someone has probably developed a module for it

Your program can adjust the way the interpreter works



Python syntax

Extremely simplified syntax

Nearly devoid of special characters

Intended to be nearly English



Dynamic types

It is the responsibility of the programmer

Still “strongly typed”

INDUSTRIAL USE OF PYTHON



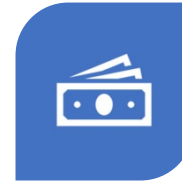
**WEB
DEVELOPMENT**



GAMES



GRAPHICS



FINANCIAL



SCIENCE



**ELECTRONIC
DESIGN
AUTOMATION**



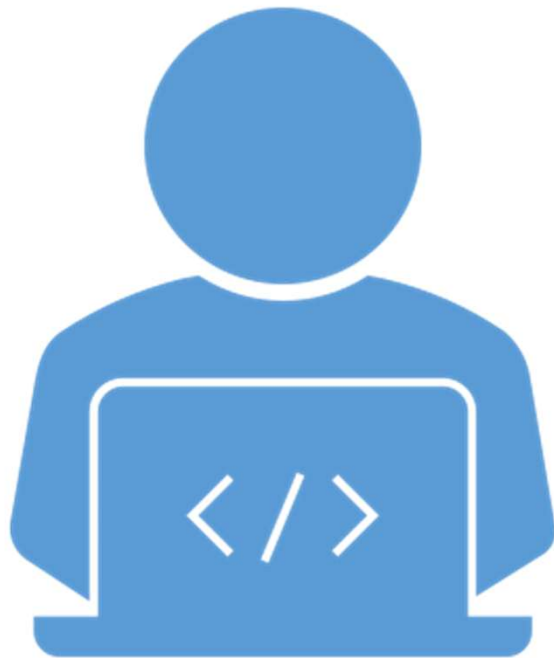
EDUCATION



**BUSINESS
SOFTWARE**

INTERACTIVE MODE

Running



- We assume that you have the Python interpreter set in PATH variable. Now, try to run this program as follows-
- On Linux: `$ python test.py`

This produces the following result-

- Hello, Python!
- On Windows
- `C:\Python37>Python test.py`

This produces the following result-

- Hello, Python!

Python Identifiers



An identifier starts with a letter A to Z or a to z or an underscore (_) followed by zero or more letters, underscores and digits (0 to 9).



Python does not allow punctuation characters such as @, \$, and % within identifiers. Python is a case sensitive programming language

Naming conventions for python identifiers

- Class names start with an uppercase letter. All other identifiers start with a lowercase letter.
- Starting an identifier with a single leading underscore indicates that the identifier is private.
- Starting an identifier with two leading underscores indicates a strong private identifier.
- If the identifier also ends with two trailing underscores, the identifier is a language defined special name.

Key Words

and	exec	Not
as	finally	or
assert	for	pass
break	from	print
class	global	raise
continue	if	return
def	import	try
del	in	while
elif	is	with
else	lambda	yield
except		

Lines and Indentation

- Python does not use braces({}) to indicate blocks of code for class and function definitions or flow control. Blocks of code are denoted by |
- if True:
- print ("True")
- else:
- print ("False") ine indentation, which is rigidly enforced.

Multi-Line Statements

- Statements in Python typically end with a new line. Python, however, allows the use of the line continuation character (\) to denote that the line should continue.

```
total = item_one + \  
        item_two + \  
        item_three
```

The statements contained within the [], {}, or () brackets do not need to use the line continuation character. For example-

```
days = ['Monday', 'Tuesday', 'Wednesday',  
        'Thursday', 'Friday']
```

Quotation in Python

- Python accepts single ('), double (") and triple (''' or """) quotes to denote string literals, as long as the same type of quote starts and ends the string.

Comments in Python

- A hash sign (#) that is not inside a string literal is the beginning of a comment. All characters after the #, up to the end of the physical line, are part of the comment and the Python interpreter ignores them.
- # First comment
- `print ("Hello, Python!")` # second comment

Waiting for the User

- `input("\Press the enter key to exit.")`



Multiple Statements on a Single Line

- The semicolon (;) allows multiple statements on a single line given that no statement starts a new code block.
- Example

Suites

- Groups of individual statements, which make a single code block are called suites in Python. Compound or complex statements, such as if, while, def, and class require a header line and a suite
- Header lines begin the statement (with the keyword) and terminate with a colon (:) and are followed by one or more lines which make up the suite.

if expression :

 suite

elif expression :

 suite

else :

 suite

Your First Python Program

- At the prompt (>>>) type:

```
print "Game Over"
```

- Press [Enter]
- Very straightforward
 - You could have guessed what this does without knowing Python!

PYTHON-VARIABLES

BY AMAR PANCHAL

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Assigning Values to Variables

- Python variables do not need explicit declaration to reserve memory space. The declaration happens automatically when you assign a value to a variable. The equal sign (=) is used to assign values to variables.
- Python allows you to assign a single value to several variables simultaneously.
- EXAMPLE
- Special case also
 - `a, b, c = 24, 36, "AMAR"`

Standard Data Types

- Python has five standard data types-
 - Numbers
 - Boolean
 - String
 - List
 - Tuple
 - Dictionary
 - Set

Numbers

- int (signed integers)
- float (floating point real values)
- complex (complex numbers)

Lists

- A list contains items separated by commas and enclosed within square brackets ([]). To some extent, lists are similar to arrays in C. One of the differences between them is that all the items belonging to a list can be of different data type.
- The values stored in a list can be accessed using the slice operator ([] and [:]) with indexes starting at 0 in the beginning of the list and working their way to end -1. The plus (+) sign is the list concatenation operator, and the asterisk (*) is the repetition operator.

```
list = [ 'amp', 4624 , 3.14, 'python', 0.07 ]
```

```
tinylis = [420, 'abcd']
```

```
print (list)
```

```
print (list[0])
```

```
print (list[1:3])
```

```
print (list[2:])
```

```
print (tinylis * 2)
```

```
print (list + tinylis)
```

LIST

Function	Description
<code>all()</code>	Return True if all elements of the list are true (or if the list is empty).
<code>any()</code>	Return True if any element of the list is true. If the list is empty, return False.
<code>enumerate()</code>	Return an enumerate object. It contains the index and value of all the items of list as a tuple.
<code>len()</code>	Return the length (the number of items) in the list.
<code>list()</code>	Convert an iterable (tuple, string, set, dictionary) to a list.
<code>max()</code>	Return the largest item in the list.
<code>min()</code>	Return the smallest item in the list
<code>sorted()</code>	Return a new sorted list (does not sort the list itself).
<code>sum()</code>	Return the sum of all elements in the list.

- `fruits = ['orange', 'apple', 'pear', 'banana', 'kiwi', 'apple', 'banana']`

```
>>> fruits.count('apple')
>>> fruits.count('tangerine')
>>> fruits.index('banana')
>>> fruits.index('banana', 4)
>>> fruits.reverse()
>>> fruits
>>> fruits.append('grape')
>>> fruits
```

Tuples

- A tuple consists of a number of values separated by commas. Unlike lists, however, tuples are enclosed within parenthesis.
- The main difference between lists and tuples are – Lists are enclosed in brackets ([]) and their elements and size can be changed, while tuples are enclosed in parentheses (()) and cannot be updated. Tuples can be thought of as **read-only** lists.

```
tuples = ( 'amp', 4624 , 3.14, 'python', 0.07 )
```

```
tinylis = (420, 'abcd')
```

```
print (list)
```

```
print (list[0])
```

```
print (list[1:3])
```

```
print (list[2:])
```

```
print (tinylis * 2)
```

```
print (list + tinylis)
```

del

The `del` statement

CODE:

```
>>> a = [-1, 1, 66.25, 333, 333, 1234.5]
```

```
>>> del a[0]
```

```
>>> a
```

```
>>> del a[2:4]
```

```
>>> a
```

```
>>> del a[:]
```

```
>>> a
```

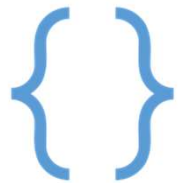
```
>>> del a
```

```
>>> print(a)
```

Dictionary

- Python's dictionaries are kind of hash-table type. They work like associative arrays or hashes found in Perl and consist of key-value pairs. A dictionary key can be almost any Python type, but are usually numbers or strings. Values, on the other hand, can be any arbitrary Python object.
- Dictionaries are enclosed by curly braces ({ }) and values can be assigned and accessed using square braces ([]).
- `dict = {}`
- `dict['one'] = "This is one"`
- `dict[2] = "This is two"`
- `tinydict = {'name': 'john', 'code': 6734, 'dept': 'sales'}`
- `print (dict['one'])`
- `print (dict[2])`
- `print (tinydict)`
- `print (tinydict.keys())`
- `print (tinydict.values())`

Method	Description
<code>clear()</code>	Remove all items from the dictionary.
<code>copy()</code>	Return a shallow copy of the dictionary.
<code>fromkeys(seq[, v])</code>	Return a new dictionary with keys from seq and value equal to v(defaults to None).
<code>get(key[,d])</code>	Return the value of key. If key doesnot exit, return d (defaults to None).
<code>items()</code>	Return a new view of the dictionary's items (key, value).
<code>keys()</code>	Return a new view of the dictionary's keys.
<code>pop(key[,d])</code>	Remove the item with key and return its value or d if key is not found. If d is not provided and key is not found, raises <code>KeyError</code> .
<code>popitem()</code>	Remove and return an arbitrary item (key, value). Raises <code>KeyError</code> if the dictionary is empty.
<code>setdefault(key[,d])</code>	If key is in the dictionary, return its value. If not, insert key with a value of d and return d (defaults to None).
<code>update([other])</code>	Update the dictionary with the key/value pairs from other, overwriting existing keys.
<code>values()</code>	Return a new view of the dictionary's values



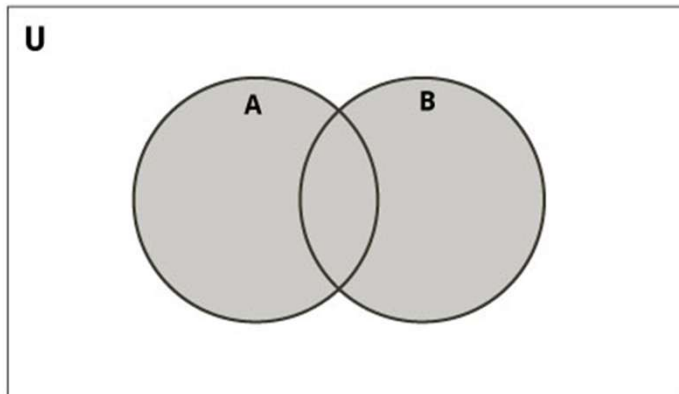
Set

- A set is a collection which is unordered and unindexed. In Python sets are written with curly brackets.



Python Set Methods	
Method	Description
<code>add()</code>	Add an element to a set
<code>clear()</code>	Remove all elements form a set
<code>copy()</code>	Return a shallow copy of a set
<code>difference()</code>	Return the difference of two or more sets as a new set
<code>difference_update()</code>	Remove all elements of another set from this set
<code>discard()</code>	Remove an element from set if it is a member. (Do nothing if the element is not in set)
<code>intersection()</code>	Return the intersection of two sets as a new set
<code>intersection_update()</code>	Update the set with the intersection of itself and another
<code>isdisjoint()</code>	Return True if two sets have a null intersection
<code>issubset()</code>	Return True if another set contains this set
<code>issuperset()</code>	Return True if this set contains another set
<code>pop()</code>	Remove and return an arbitrary set element. Raise <code>KeyError</code> if the set is empty
<code>remove()</code>	Remove an element from a set. If the element is not a member, raise a <code>KeyError</code>
<code>symmetric_difference()</code>	Return the symmetric difference of two sets as a new set
<code>symmetric_difference_update()</code>	Update a set with the symmetric difference of itself and another
<code>union()</code>	Return the union of sets in a new set
<code>update()</code>	Update a set with the union of itself and others

Set Union



initialize A and B

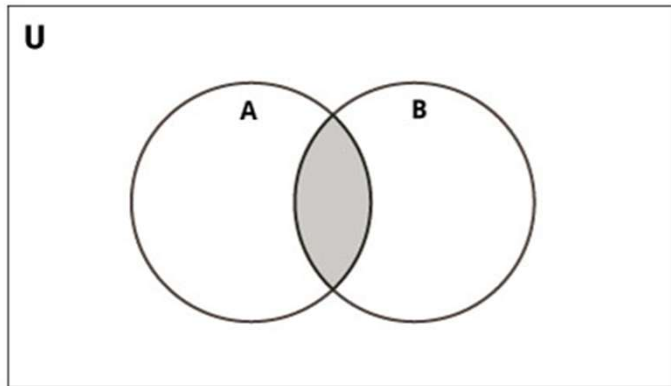
$A = \{1, 2, 3, 4, 5\}$

$B = \{4, 5, 6, 7, 8\}$

use | operator

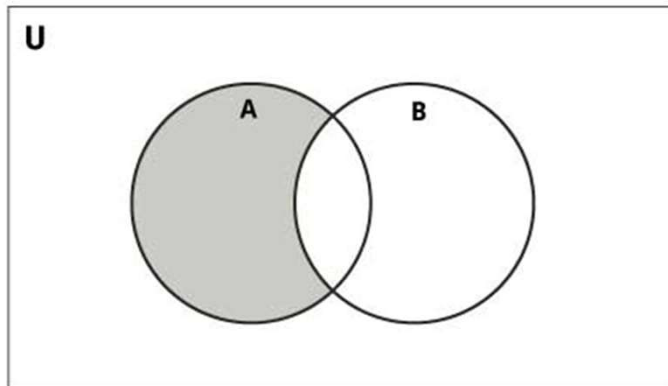
`print(A | B)`

Set Intersection



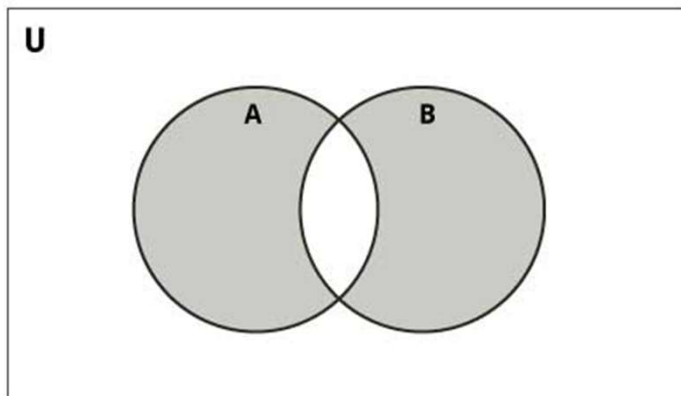
- # initialize A and B
- A = {1, 2, 3, 4, 5}
- B = {4, 5, 6, 7, 8}
- # use & operator#
- print(A & B)

Set Difference



- # initialize A and B
- $A = \{1, 2, 3, 4, 5\}$
- $B = \{4, 5, 6, 7, 8\}$
- `print(A - B)`

Set Symmetric Difference



- # initialize A and B
- A = {1, 2, 3, 4, 5}
- B = {4, 5, 6, 7, 8}
- print(A ^ B)

Backslash notation	Hexadecimal character	Description
\a	0x07	Bell or alert
\b	0x08	Backspace
\cx		Control-x
\C-x		Control-x
\e	0x1b	Escape
\f	0x0c	Formfeed
\M-\C-x		Meta-Control-x
\n	0x0a	Newline
\nnn		Octal notation, where n is in the range 0-7
\r	0x0d	Carriage return
\s	0x20	Space
\t	0x09	Tab
\v	0x0b	Vertical tab
\x		Character x
\xnn		Hexadecimal notation, where n is in the range 0-9, a-f, or A-F

Data Type Conversion

Function	Description
<code>int(x [,base])</code>	Converts x to an integer. The base specifies the base if x is as string.
<code>float(x)</code>	Converts x to a floating-point number.
<code>complex(real[,imag])</code>	Creates a complex number.
<code>str(x)</code>	Converts object x to a string representation.
<code>repr(x)</code>	Converts object x to an expression string.
<code>eval(str)</code>	Evaluates a string and returns an object.
<code>tuple(s)</code>	Converts s to a tuple.
<code>list(s)</code>	Converts s to a list.
<code>set(s)</code>	Converts s to a set.
<code>dict(d)</code>	Creates a dictionary. d must be a sequence of (key,value) tuples.
<code>frozenset(s)</code>	Converts s to a frozen set.
<code>chr(x)</code>	Converts an integer to a character.
<code>unichr(x)</code>	Converts an integer to a Unicode character.
<code>ord(x)</code>	Converts a single character to its integer value.
<code>hex(x)</code>	Converts an integer to a hexadecimal string
<code>oct(x)</code>	Converts an integer to an octal string.

PYTHON-OPERATORS

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Types of Operator

Arithmetic Operators

Comparison (Relational)
Operators

Assignment Operators

Logical Operators

Bitwise Operators

Membership Operators

Identity Operators

Arithmetic Operators

Operator	Description	Example
+ Addition	Adds values on either side of the operator.	$a + b = 31$
- Subtraction	Subtracts right hand operand from left hand operand.	$a - b = -11$
* Multiplication	Multiplies values on either side of the operator	$a * b = 210$
/ Division	Divides left hand operand by right hand operand	$b / a = 2.1$
% Modulus	Divides left hand operand by right hand operand and returns remainder	$b \% a = 1$
** Exponent	Performs exponential (power) calculation on operators	$a ** b = 10 \text{ to the power } 20$
//	Floor Division - The division of operands where the result is the quotient in which the digits after the decimal point are removed. But if one of the operands is negative, the result is floored, i.e., rounded away from zero (towards negative infinity):	$9 // 2 = 4$ and $9.0 // 2.0 = 4.0$, $-11 // 3 = -4$, $-11.0 // 3 = -4.0$

Comparison Operators

Operator	Description	Example
==	If the values of two operands are equal, then the condition becomes true.	(a == b) is not true.
!=	If values of two operands are not equal, then condition becomes true.	(a != b) is true.
>	If the value of left operand is greater than the value of right operand, then condition becomes true.	(a > b) is not true.
<	If the value of left operand is less than the value of right operand, then condition becomes true.	(a < b) is true.
>=	If the value of left operand is greater than or equal to the value of right operand, then condition becomes true.	(a >= b) is not true.
<=	If the value of left operand is less than or equal to the value of right operand, then condition becomes true.	(a <= b) is true.

Assignment Operators

Operator	Description	Example
=	Assigns values from right side operands to left side operand	$c = a + b$ assigns value of $a + b$ into c
$+=$ Add AND	It adds right operand to the left operand and assign the result to left operand	$c += a$ is equivalent to $c = c + a$
$-=$ Subtract AND	It subtracts right operand from the left operand and assign the result to left operand	$c -= a$ is equivalent to $c = c - a$
$*=$ Multiply AND	It multiplies right operand with the left operand and assign the result to left operand	$c *= a$ is equivalent to $c = c * a$
$/=$ Divide AND	It divides left operand with the right operand and assign the result to left operand	$c /= a$ is equivalent to $c = c / a$
$\% =$ Modulus AND	It takes modulus using two operands and assign the result to left operand	$c \% = a$ is equivalent to $c = c \% a$
$**=$ Exponent AND	Performs exponential (power) calculation on operators and assign value to the left operand	$c ** = a$ is equivalent to $c = c ** a$
$// =$ Floor Division	It performs floor division on operators and assign value to the left operand	$c // = a$ is equivalent to $c = c // a$

Bitwise Operators

Operator	Description	Example
& Binary AND	Operator copies a bit, to the result, if it exists in both operands	(a & b) (means 0000 1100)
Binary OR	It copies a bit, if it exists in either operand.	(a b) = 61 (means 0011 1101)
^ Binary XOR	It copies the bit, if it is set in one operand but not both.	(a ^ b) = 49 (means 0011 0001)
~ Binary Ones Complement	It is unary and has the effect of 'flipping' bits.	(~a) = -61 (means 1100 0011 in 2's complement form due to a signed binary number.
<< Binary Left Shift	The left operand's value is moved left by the number of bits specified by the right operand.	a << = 240 (means 1111 0000)
>> Binary Right Shift	The left operand's value is moved right by the number of bits specified by the right operand.	a >> = 15 (means 0000 1111)

Logical Operators

Operator	Description	Example
and Logical AND	If both the operands are true then condition becomes true.	(a and b) is False.
or Logical OR	If any of the two operands are non-zero then condition becomes true.	(a or b) is True.
not Logical NOT	Used to reverse the logical state of its operand.	Not(a and b) is True.

Membership Operators

- Python's membership operators test for membership in a sequence, such as strings, lists, or tuples.

Operator	Description	Example
in	Evaluates to true if it finds a variable in the specified sequence and false otherwise.	x in y, here in results in a 1 if x is a member of sequence y.
not in	Evaluates to true if it does not finds a variable in the specified sequence and false otherwise.	x not in y, here not in results in a 1 if x is not a member of sequence y.

Identity Operators

Identity operators compare the memory locations of two objects



Operator	Description	Example
is	Evaluates to true if the variables on either side of the operator point to the same object and false otherwise.	x is y, here is results in 1 if id(x) equals id(y).
is not	Evaluates to false if the variables on either side of the operator point to the same object and true otherwise.	x is not y, here is not results in 1 if id(x) is not equal to id(y).

Operators Precedence

S.No.	Operator & Description
1	** Exponentiation (raise to the power)
2	~ + - Ccomplement, unary plus and minus (method names for the last two are +@ and -@)
3	* / % // Multiply, divide, modulo and floor division
4	+ - Addition and subtraction
5	>> << Right and left bitwise shift
6	& Bitwise 'AND'

Operators Precedence

S.No.	Operator & Description
7	^ Bitwise exclusive `OR' and regular `OR'
8	<= < > >= Comparison operators
9	<> == != Equality operators
10	= %= /= //= -= += *= **= Assignment operators
11	is is not Identity operators
12	in not in Membership operators
13	not or and Logical operators

PYTHON CONDITIONAL STATEMENT & LOOPS

BY
AMAR PANCHAL

IF

if Statements

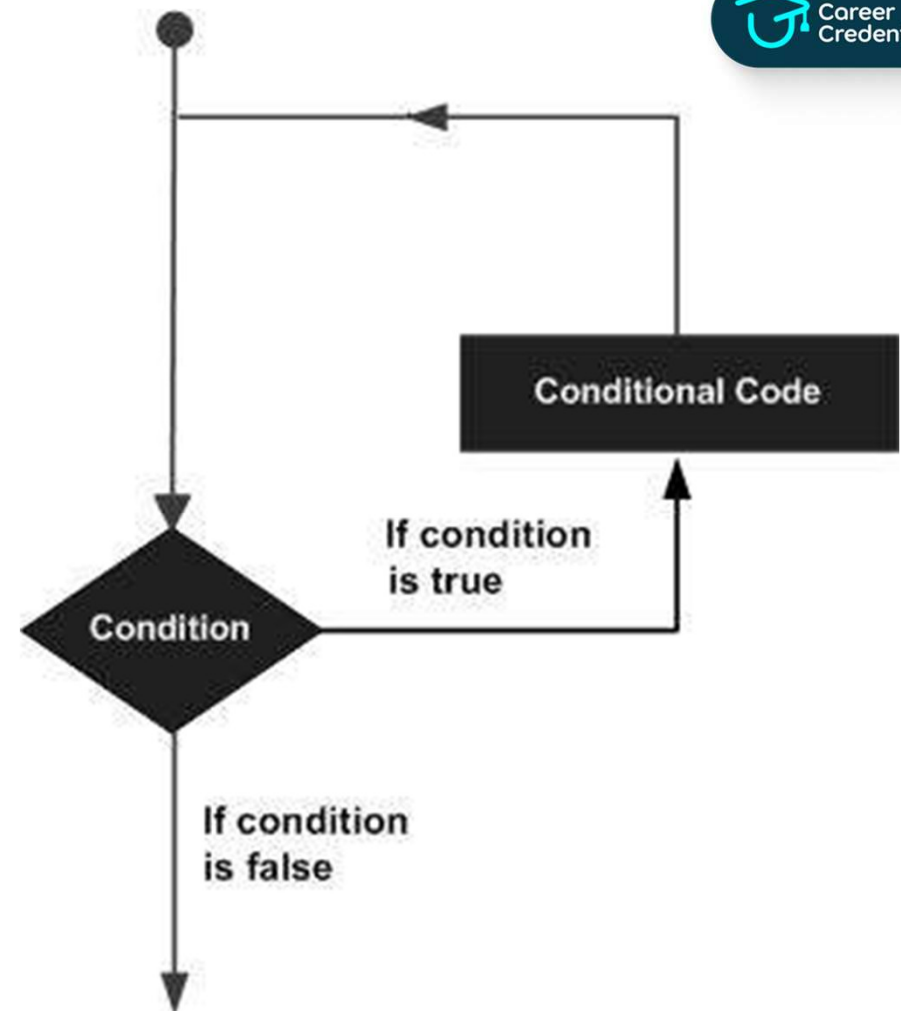
It consists of a Boolean expression which results is either TRUE or FALSE followed by one or more statements.

```
if expression:  
    #execute your code  
  
if expression:  
    #execute your code  
else:  
    #execute your code
```

```
if expression:  
    #execute your code  
elif expression:  
    #execute your code  
else:  
    #execute your code
```


Loops

- A loop statement allows us to execute a statement or group of statements multiple times.
- The following diagram illustrates a loop statement –
-



While

- Syntax

while expression:
 statement(s)

Ex

```
count = 0  
while (count < 5):  
    print ('The count is:',  
    count)  
    count = count + 1  
  
print ("Good bye!")
```

for Loop

- Syntax

for iterating_var in sequence:
 statements(s)

- The range() function

The built-in function range() is the right function to iterate over a sequence of numbers. It generates an iterator of arithmetic progressions.

Example

```
>>> range(5)
```

For Loop

- For i in range(5):
 print(i)
- For i in “python”:
 print(i)

Single Statement Suites

```
flag = 1  
while (flag): print ('Given flag is really true!')  
print ("Good bye!")
```

Loop Control Statements



S.No.	Control Statement & Description
1	break statement: Terminates the loop statement and transfers execution to the statement immediately following the loop.
2	continue statement: Causes the loop to skip the remainder of its body and immediately retest its condition prior to reiterating.
3	pass statement: The pass statement in Python is used when a statement is required syntactically but you do not want any command or code to execute.

else block with loops

```
no = int(input('any number: '))
numbers = [11,33,55,39,55,75,37,21,23,41,13]
for num in numbers:
    if num == no:
        print ('number found in list')
        break
else:
    print ('number not found in list')
```

```
numbers = [11,33,55,39,55,75,37,21,23,41,13]
```

```
for num in numbers:
```

```
    if num%2 == 0:
```

```
        print ('the list contains an even number')
```

```
        break
```

```
else:
```

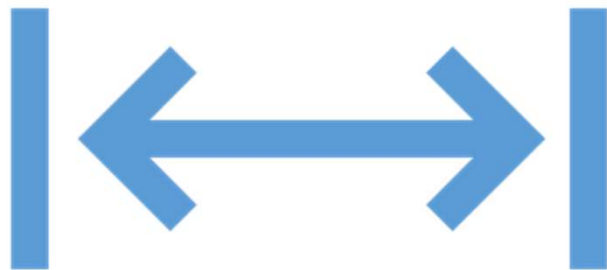
```
    print ('the list does not contain even number')
```


Python-Functions

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Defining a Function



- Function blocks begin with the keyword **def** followed by the function name and parentheses (())

```
def functionname( parameters ):  
    "function_docstring"  
    function_suite  
    return [expression]
```

Function Arguments



- Required arguments
- Keyword arguments
- Default arguments
- Variable-length arguments

Scope of Variables

- **Global variables**
- **Local variables**

```
total = 0
```

```
def sum( arg1, arg2 ):
    total = arg1 + arg2; # Here total is local variable.
    print ("Inside the function local total : ", total)
    return total
```

```
sum( 10, 20 )
print ("Outside the function global total : ", total )
```


PYTHON –OOPS

BY AMAR PANCHAL

Creating Classes



```
class ClassName:  
    'Optional class documentation string'  
    class_suite
```


Creating Instance Objects



Accessing Attributes

- The `getattr(obj, name[, default])` – to access the attribute of object.
- The `hasattr(obj,name)` – to check if an attribute exists or not.
- The `setattr(obj,name,value)` – to set an attribute. If attribute does not exist, then it would be created.
- The `delattr(obj, name)` – to delete an attribute.

Importing



Built-In Class Attributes

- **__dict__** – Dictionary containing the class's namespace.
- **__doc__** – Class documentation string or none, if undefined.
- **__name__** – Class name.
- **__module__** – Module name in which the class is defined. This attribute is "**__main__**" in interactive mode.
- **__bases__** – A possibly empty tuple containing the base classes, in the order of their occurrence in the base class list.

Class Inheritance



```
class SubClassName (ParentClass1[, ParentClass2, ...]):  
    'Optional class documentation string'  
    class_suite
```

check a relationships

- The **issubclass(sub, sup)** boolean function returns True, if the given subclass **sub** is indeed a subclass of the superclass **sup**.
- The **isinstance(obj, Class)** boolean function returns True, if *obj* is an instance of class *Class* or is an instance of a subclass of Class

Overriding Methods

S.No.	Method, Description & Sample Call
1	<code>__init__ (self [,args...])</code> Constructor (with any optional arguments) Sample Call : <code>obj = className(args)</code>
2	<code>__del__(self)</code> Destructor, deletes an object Sample Call : <code>del obj</code>
3	<code>__repr__(self)</code> Evaluatable string representation Sample Call : <code>repr(obj)</code>
4	<code>__str__(self)</code> Printable string representation Sample Call : <code>str(obj)</code>
5	<code>__cmp__(self, x)</code> Object comparison Sample Call : <code>cmp(obj, x)</code>

Data Hiding

- The __ sign

PYTHON-EXCEPTIONS

BY
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EXCEPTION HANDLING

- **Exception & Assertions**

Handling an exception

try:

You do your operations here

.....

except ExceptionI:

If there is ExceptionI, then execute this block.

except ExceptionII:

If there is ExceptionII, then execute this block.

.....

else:

If there is no exception then execute this block.

The except Clause with Multiple Exceptions

try:

You do your operations here

.....

```
except(Exception1[,  
Exception2[,...ExceptionN]]):
```

If there is any exception from the given
exception list,

then execute this block.

.....

else:

If there is no exception then execute this
block

The try-finally Clause

try:

You do your operations here;

.....

Due to any exception, this may be skipped.

finally:

This would always be executed.

.....

Argument of an Exception

try:

You do your operations here

.....

except ExceptionType as Argument:

You can print value of Argument here...

Raising an Exception

- Syntax
`raise [Exception [, args [, traceback]]]`

User-Defined Exceptions

```
class MyError(Exception):  
    def __init__(self, value):  
        self.value = value  
    def __str__(self):  
        return self.value  
  
try:  
    raise MyError(2*2)  
except MyError, e:  
    print("My exception occurred:", e.value)
```


Assertions in Python

- The syntax for assert is –
`assert Expression[, Arguments]`

PYTHON-FILES

BY
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The open Function

- `file object = open(file_name [, access_mode][, buffering])`

modes

MODE	
r	
r+	
w	
w+	
a	
a+	
rb	
wb	
ab	
rb+	
wb+	
ab+	

The file Object Attributes

S.No.	Attribute & Description
1	file.closed Returns true if file is closed, false otherwise.
2	file.mode Returns access mode with which file was opened.
3	file.name Returns name of the file.

code

```
f = open("MyFile", "w")  
print ("Name of the file: ", f.name)  
print ("Closed or not : ", f.closed)  
print ("Opening mode : ", f.mode)  
f.close()
```

The close()

- `fileObject.close()`

Reading and Writing Files

- The **read()** method reads a string from an open file.
- `fileObject.read([count]);`

Reading and Writing Files

- The **write()** method writes any string to an open file.

```
fileObject.write(string);  
f = open("MyFile.txt", "w")  
f.write( "file operation writing  
process")  
f.close()
```

File Positions

tell()

seek(offset[, from])

Renaming and Deleting Files



- Python **os** module provides methods that help you perform file-processing operations, such as renaming and deleting files. To use this module, you need to import it.

rename()	<code>os.rename(current_file_name, new_file_name)</code>
remove()	<code>os.remove(file_name)</code>

Dealing With Directories

Command	Use
<code>os.mkdir("newdir")</code>	
<code>os.chdir("newdir")</code>	
<code>os.getcwd()</code>	
<code>os.rmdir("dirname")</code>	

Python Iterators, Generators, Decorators

BY
AMAR PANCHAL

What are iterators in Python?

- **Iterators are objects that can be iterated upon.**
- Iterator in Python is simply an [object](#) that can be iterated upon. An object which will return data, one element at a time.
- Technically speaking, Python iterator object must implement two special methods, `__iter__()` and `__next__()`, collectively called the iterator protocol.

CODE

- `my_list = [10, 20, 30, 40]`
- `my_iter = iter(my_list)`
- `print(next(my_iter))`
- `print(next(my_iter))`
- `print(my_iter.__next__())`
- `print(my_iter.__next__())`

Building Your Own Iterator in Python

- We just have to implement the methods `__iter__()` and `__next__()`.

CODE

```
class PowTwo:
```

```
    def __init__(self, max = 0):  
        self.max = max  
        self.n = 0
```

```
    def __next__(self):  
        if self.n <= self.max:  
            result = 2 **  
  
            self.n += 1  
            return result
```

```
    self.n
```

```
        else:  
            raise StopIteration
```

What are generators in Python?

- Python generators are a simple way of creating iterators.
- Simply speaking, a generator is a function that returns an object (iterator) which we can iterate over (one value at a time).
- It is fairly simple to create a generator in Python. It is as easy as defining a normal function with yield statement instead of a return statement.
- If a function contains at least one yield statement (it may contain other yield or return statements), it becomes a generator function. Both yield and return will return some value from a function.
- The difference is that, while a return statement terminates a function entirely, yield statement pauses the function saving all its states and later continues from there on successive calls.

Differences between Generator function and a Normal function

- Generator function contains one or more yield statement.
- When called, it returns an object (iterator) but does not start execution immediately.
- Methods like `__iter__()` and `__next__()` are implemented automatically. So we can iterate through the items using `next()`.
- Once the function yields, the function is paused and the control is transferred to the caller.
- Local variables and their states are remembered between successive calls.
- Finally, when the function terminates, `StopIteration` is raised automatically on further calls.

CODE

```
# A simple generator function
def my_gen():
    n = 1
    print('This is printed first')
    yield n
    n += 1
    print('This is printed
second')
    yield n
    n += 1
    print('This is printed at last')
    yield n
```

String

```
def rev_str(my_str):  
    length = len(my_str)  
    for i in range(length - 1, -1, -1):
```

Python Generator Expression

- `my_list = [1, 3, 6, 10]`
- `a = (x**2 for x in my_list)`

Why generators are used in Python?

- 1. Easy to Implement**
- 2. Memory Efficient**
- 3. Represent Infinite Stream**

Python Decorators

- Python has an interesting feature called **decorators** to add functionality to an existing code.
- This is also called **metaprogramming** as a part of the program tries to modify another part of the program at compile time.

Functions are Objects

```
def first(msg):  
    print(msg)  
  
first("Hello")  
second = first  
second("Hello")
```

Functions inside Functions

```
def f():  
  
    def g():  
        print("Hi, it's me 'g'")#4  
        print("Thanks for calling me")#5  
  
    print("This is the function 'f'")#1  
    print("I am calling 'g' now:")#2  
    g()#3
```

```
f()
```

Functions as Parameters

```
def g():  
    print("Hi, it's me 'g'")  
    print("Thanks for calling me")
```

```
def f(func):  
    print("Hi, it's me 'f'")  
    print("I will call 'func' now")  
    func()
```

```
f(g)
```

```
def g():  
    print("Hi, it's me 'g'")  
    print("Thanks for calling me")  
  
def f(func):  
    print("Hi, it's me 'f'")  
    print("I will call 'func' now")  
    func()  
    print("func's real name is " + func.__name__)
```

f(g)

Functions returning Functions

```
def f(x):  
    def g(y):  
        return y + x + 3  
    return g
```

```
nf1 = f(1)  
nf2 = f(3)
```

```
print(nf1(1))  
print(nf2(1))
```

A Simple Decorator



```
def our_decorator(func):  
    def function_wrapper(x):  
        print("Before calling " + func.__name__)  
        func(x)  
        print("After calling " + func.__name__)  
    return function_wrapper  
  
def foo(x):  
    print("Hi, foo has been called with " + str(x))  
  
print("We call foo before decoration:")  
foo("Hi")  
  
print("We now decorate foo with f:")  
foo = our_decorator(foo)  
  
print("We call foo after decoration:")  
foo(42)
```


The Usual Syntax for Decorators in Python

- The decoration in Python is usually not performed in the way we did it in our previous example, even though the notation `foo = our_decorator(foo)` is catchy and easy to grasp. This is the reason, why we used it! You can also see a design problem in our previous approach. "foo" existed in the same program in two versions, before decoration and after decoration.
- We will do a proper decoration now. The decoration occurs in the line before the function header. The "@" is followed by the decorator function name.
- We will rewrite now our initial example. Instead of writing the statement
- `foo = our_decorator(foo)`
- we can write
- `@our_decorator`

```
def our_decorator(func):  
    def function_wrapper(x):  
        print("Before calling: " + func.__name__)  
        func(x)  
        print("After calling : " + func.__name__)  
    return function_wrapper  
  
@our_decorator  
def foo(x):  
    print("Hi, foo has been called with " + str(x))  
  
foo("Hi")
```